A11 Box – Build and Test, Stage by Stage · United Control Box, wiring rev 0.7 · 2026-10-29 · 10 stages

Wire one stage, then pass all its tests before the next: each stage only adds parts the earlier stages already proved, so a fault shows up where it was made. Columns, cut lengths, bands and end codes as in A11_Wire_List.pdf (fer = ferrule, fer·own = own point on the distribution block, 2×+N = twin ferrule with wire N, fork(+N) = fork lug (2 under one screw with N), ring = ring lug, solder = GX16 cup, crimp = HDC-HA contact, core = supply cable core). **Tests:** LIVE = powered, dead = QF0 OFF and the cord unplugged, Ω / beep meter. Values from the § 7.1 / § 7.2 procedures.

⚠ **Between stages:** QF0 (main breaker) OFF and the cord unplugged; prove it dead with the meter (Z1 LINE L ↔ N, L ↔ PE) before you touch. **Live tests:** one hand, insulated probes, a CAT III meter, nobody else at the box. **A test fails:** stop, fix it and repeat that stage's tests before the next stage. Number sleeve and colour band on both ends; ferrule every stranded end; max. 2 conductors per terminal (R1).

	No.	From	To	Wire mm²	Band	Cut	Ends (from / to)	Function · ⚠ check before / while wiring
Stage 1 · Protective earth – 11 wires, 12.80 m								
⚠ **CHECK FIRST:** All parts and ducts mounted, no other wire yet. Scrape the paint under the body and plate studs, serrated washer under the lug. PE is G/Y only, never plain yellow. After the group: every PE point beeps to the earth bar.								
☐	**138**	Earth bar 1	Earth bar 2	4 G/Y		**0.60**	ring M6 / ring M6	link (both bars sit on insulators)
☐	**139**	Enclosure body (earth stud)	Earth bar 1	4 G/Y		**0.80**	ring M6 / ring M6	NEW: the body had no PE conductor ⚠ **bare metal under the lug**
☐	**141**	Backplate	Earth bar 1	4 G/Y		**0.45**	ring M6 / ring M6	⚠ **bare metal under the lug**
☐	**142**	Door strap	Earth bar 1	6 braided strap		**1.40**	ring M6 / ring M6	door loop
☐	**143**	EMI filter GND	Earth bar 1	4 G/Y		**1.60**	fer / ring M6	
☐	**144**	VFD PEc	Earth bar 1	2.5 G/Y		**1.15**	fer / ring M6	
☐	**145**	PSU 48V PE	Earth bar 2	2.5 G/Y		**1.30**	fork / ring M6	
☐	**146**	PSU 24V PE	Earth bar 2	2.5 G/Y		**1.15**	fork / ring M6	
☐	**147**	DIN-rail earth stud	Earth bar 2	2.5 G/Y		**1.65**	ring M6 / ring M6	
☐	**148**	PSU 24V -V2	Earth bar 2	2.5 G/Y		**1.20**	fork+169 / ring M6	PELV bond: the only 0 V to PE bond ⚠ **must beep 0 V ↔ PE in § 7.1 (dead tests)**
☐	**149**	Schuko PE	Earth bar 2	2.5 G/Y		**1.50**	fer / ring M6	NEW: missing in rev 0.5 (pump was unearthed)
		TEST 1 – pass all before stage 2						**You must see**
☐	1.1	dead Each PE point (body and plate studs, door, DIN rail, Z1 (EMI filter) GND, VFD (spindle drive) PEc, G1 (48 V PSU) / G2 (24 V PSU) PE, Schuko PE) ↔ earth bar						**beep, ≈ 0 Ω**
☐	1.2	dead Earth bar 1 ↔ earth bar 2 (138); door stud ↔ body with the door fully open (142)						**beep**
☐	1.3	dead G2 (24 V PSU) −V ↔ earth bar (148, the PELV (earthed 24 V) bond)						**beep**
Stage 2 · Supply, EMI filter and 24 V PSU – 11 wires, 9.75 m								
⚠ **CHECK FIRST:** Cord unplugged. Read the labels: Z1 (EMI filter) LINE in from QF0 (main breaker), LOAD out. F1 (control fuse) / F2 (signal fuse) carriers open while wiring. Supply cable last: PE (140) first, then L (101) and N (102).								
☐	**140**	Mains (gland) PE	Earth bar 1	4 G/Y		**0.65**	core / ring M6	⚠ **connect last, after § 7.1 (dead tests)**
☐	**101**	Mains (gland) L	Main CB 1	2.5 black		**1.50**	core / fer	Mains L ⚠ **connect the supply cable last, after the § 7.1 (dead tests) dead tests**
☐	**102**	Mains (gland) N	Main CB 3	2.5 blue		**1.55**	core / fer	Mains N ⚠ **connect last, after § 7.1 (dead tests)**
☐	**103**	Main CB 2	EMI filter LIN	4 black		**0.30**	fer / fer	⚠ **Z1 (EMI filter) LINE side (from QF0 (main breaker)); do not swap LINE / LOAD**
☐	**104**	Main CB 4	EMI filter NIN	4 blue		**0.35**	fer / fer	⚠ **Z1 (EMI filter) LINE side; do not swap LINE / LOAD**
☐	**106**	EMI filter NOUT	N busbar XN	4 blue		**0.90**	fer / fer·own	N in
☐	**111**	EMI filter LOUT	CB-24V 1	4 black		**0.55**	2×+105 / fer	24 V PSU L (unswitched)
☐	**112**	CB-24V 2	PSU 24V L	1 yellow	**black**	**1.00**	2×+130 / fork	
☐	**113**	N busbar XN	PSU 24V N	1 yellow	**blue**	**0.85**	fer·own / fork	
☐	**162**	PSU 24V +V1	F1 in	2.5 brown	**dk blue**	**1.10**	fork / fer	Control fuse (short, 2.5 mm²: unfused PSU side)
☐	**164**	PSU 24V +V2	F2 in	2.5 brown	**dk blue**	**1.00**	fork / fer	Signal fuse (short, 2.5 mm²: unfused PSU side)
		TEST 2 – pass all before stage 3						**You must see**
☐	2.1	dead QF0 (main breaker) OFF: Z1 (EMI filter) LOUT ↔ Z1 NOUT						**open, or high (filter bleed)**
☐	2.2	dead QF0 (main breaker) ON, cord still unplugged: plug L ↔ PE and plug N ↔ PE						**open**
☐	2.3	dead G2 (24 V PSU) +V ↔ G2 −V						**not 0 Ω**
☐	2.4	**LIVE** Plug in; QF1–QF3 OFF, F1 (control fuse) / F2 (signal fuse) open; QF0 (main breaker) ON, QF4 (24 V PSU + coil breaker) ON: G2 (24 V PSU) +V ↔ G2 −V						**23–25 V DC, +V positive (else adjust G2 (24 V PSU))**
☐	2.5	**LIVE** Close F1 (control fuse) and F2 (signal fuse): F1 out ↔ G2 (24 V PSU) −V and F2 out ↔ G2 −V						**23–25 V DC each**
☐	2.6	**LIVE** Open F1 (control fuse): F1 out ↔ G2 (24 V PSU) −V; close F1 again						**0 V**
Stage 3 · K1 self-hold: ON / OFF / E-stop (24 V) – 7 wires, 7.80 m								
⚠ **CHECK FIRST:** K1 (latch relay) in its socket. Door wires 201–205 through the spiral-wrapped hinge loop. K1 pole 4 → KM1 (130 / 131) is not wired yet, so only K1 moves in this stage.								
☐	**201**	F1 out	E-stop 1	1 yellow	**dk blue**	**1.90**	2×+220 / 2×+223	Chain; door loop
☐	**202**	E-stop 2	OFF 1	1 yellow	**dk blue**	**0.35**	fer / fer	Chain; door loop
☐	**203**	OFF 2	ON 1	1 yellow	**dk blue**	**0.25**	2×+205 / fer	Chain; door loop
☐	**204**	ON 2	K1 A1 (NEMA 13)	1 yellow	**dk blue**	**1.70**	fer / 2×+206	Start; door loop
☐	**205**	OFF 2	K1 11 (NEMA 9)	1 yellow	**dk blue**	**1.85**	2×+203 / fer	Latch; door loop
☐	**206**	K1 14 (NEMA 5)	K1 A1 (NEMA 13)	1 yellow	**dk blue**	**0.80**	fer / 2×+204	Latch
☐	**207**	PSU 24V -V1	K1 A2 (NEMA 14)	1 yellow	**bl/wh**	**0.95**	fork+167 / fer	Coil return
		TEST 3 – pass all before stage 4						**You must see**
☐	3.1	dead § 7.1 (dead tests): F1 (control fuse) out ↔ K1 (latch relay) A1, no button pushed						**open**

		No.	From	To	Wire mm²	Band	Cut	Ends (from / to)	Function · ⚠ check before / while wiring
53	☐	3.2	dead Same probes, ON held						**beep**
54	☐	3.3	dead ON held and E-stop pushed; then ON held and OFF pushed						**open, open**
55	☐	3.4	**LIVE** Push ON and release						**K1 (latch relay) pulls in and stays in (latch)**
56	☐	3.5	**LIVE** Push OFF. Then ON again and push the E-stop						**K1 (latch relay) drops out each time**
57	☐	3.6	**LIVE** E-stop pushed: push ON. Release the E-stop						**K1 (latch relay) stays off; no restart until ON**
58	**Stage 4 · Contactor KM1 (230 V coil) and yellow lamp – 8 wires, 7.55 m**								
59	⚠ **CHECK FIRST:** KM1 (contactor) coil is 230 V AC: 130 / 131 red band, 132 blue. Z1 (EMI filter) LOUT → KM1 1L1, KM1 2T1 → XL (KM1 switches L only). QF1–QF3 stay OFF: XL (switched L busbar) feeds them.								
60	☐	**130**	CB-24V 2	K1 41 (NEMA 12)	1 yellow	**red**	**0.85**	2×+112 / fer	Coil supply L (protected by QF4 C6)
61	☐	**131**	K1 44 (NEMA 8)	Contactor A2	1 yellow	**red**	**0.35**	fer / fer	Switched coil L ⚠ **leave the KM1 (contactor) A2 end off until § 7.2 (power-up)**
62	☐	**132**	N busbar XN	Contactor A1	1 yellow	**blue**	**0.65**	fer·own / fer	Coil N
63	☐	**105**	EMI filter LOUT	Contactor 1L1	4 black		**0.45**	2×+111 / fer	Unswitched L
64	☐	**109**	Contactor 2T1	L busbar XL	4 black		**0.75**	fer / fer·own	Switched L in
65	☐	**220**	F1 out	Contactor 43	1 yellow	**dk blue**	**1.05**	2×+201 / 2×+170	Yellow lamp supply
66	☐	**221**	Contactor 44	Yellow lamp +	1 yellow	**dk blue**	**1.70**	fer / fer	door loop
67	☐	**222**	K1 A2 (2nd screw) (NEMA 14)	Yellow lamp -	1 yellow	**bl/wh**	**1.75**	fer / fer	0 V to the door; door loop
68			**TEST 4 – pass all before stage 5**						**You must see**
69	☐	4.1	dead QF4 (24 V PSU + coil breaker) out ↔ KM1 (contactor) A2; then hold the K1 (latch relay) test lever						**open; beep**
70	☐	4.2	dead KM1 (contactor) A1 ↔ XN (N busbar); Z1 (EMI filter) LOUT ↔ XL (switched L busbar)						**beep; open**
71	☐	4.3	**LIVE** Push ON						**K1 (latch relay) and KM1 (contactor) pull in, yellow lamp on**
72	☐	4.4	**LIVE** XL (switched L busbar) ↔ XN (N busbar)						**207–253 V AC**
73	☐	4.5	**LIVE** Push OFF; then ON and the E-stop						**KM1 (contactor) drops, lamp off, XL (switched L busbar) ↔ XN (N busbar) ≈ 0 V**
74	**Stage 5 · Pump outlet (QF3) – 3 wires, 1.50 m**								
75	⚠ **CHECK FIRST:** Schuko PE (149) is already in from stage 1. Nothing plugged into the outlet while wiring.								
76	☐	**120**	L busbar XL	CB-PUMP 1	4 black		**0.65**	fer·own / fer	Switched L
77	☐	**121**	CB-PUMP 2	Schuko L	1 yellow	**black**	**0.40**	fer / fer	Pump
78	☐	**122**	N busbar XN	Schuko N	1 yellow	**blue**	**0.45**	fer·own / fer	Pump
79			**TEST 5 – pass all before stage 6**						**You must see**
80	☐	5.1	dead Schuko L ↔ N; L ↔ PE; N ↔ PE						**open**
81	☐	5.2	**LIVE** Push ON, QF3 (pump outlet breaker) ON: Schuko L ↔ N						**207–253 V AC**
82	☐	5.3	**LIVE** Push OFF; plug the pump in, ON, then OFF						**0 V when off; pump runs only with KM1 (contactor) in**
83	**Stage 6 · A11 controller power, fan and lamps – 10 wires, 11.80 m**								
84	⚠ **CHECK FIRST:** Dark blue band = +24 V, blue/white = 0 V. Fit 166 / 168 at F2 (signal fuse) and 167 / 169 at G2 (24 V PSU), leave their A11 (controller) ends loose until 6.2. Door wires 223–226 through the hinge loop.								
85	☐	**166**	F2 out	A11 DC24V	1 yellow	**dk blue**	**1.40**	2×+168 / fer	⚠ **A11 (controller) end off until F2 (signal fuse) out reads 23–25 V DC (§ 7.2)**
86	☐	**167**	PSU 24V -V1	A11 DCGND	1 yellow	**bl/wh**	**0.95**	fork+207 / 2×+171	
87	☐	**168**	F2 out	A11 24V	1 yellow	**dk blue**	**1.40**	2×+166 / 2×+340	Sensor/input supply ⚠ **A11 (controller) end off until F2 (signal fuse) out reads 23–25 V DC**
88	☐	**169**	PSU 24V -V2	A11 GND	1 yellow	**bl/wh**	**1.00**	fork+148 / 2×+337	
89	☐	**170**	Contactor 43	Fan +	1 yellow	**dk blue**	**1.35**	2×+220 / fer	NEW: fan behind F1 (was unfused on TB24+)
90	☐	**171**	A11 DCGND	Fan -	1 yellow	**bl/wh**	**0.85**	2×+167 / fer	
91	☐	**223**	E-stop 1	Green lamp +	1 yellow	**dk blue**	**0.50**	2×+201 / 2×+225	+24 V CONTROL (door loop); door loop
92	☐	**224**	Green lamp -	A11 Y6	1 yellow	**dk blue**	**2.00**	fer / fer	Work; door loop
93	☐	**225**	Green lamp +	Red lamp +	1 yellow	**dk blue**	**0.30**	2×+223 / fer	door loop
94	☐	**226**	Red lamp -	A11 Y5	1 yellow	**dk blue**	**2.05**	fer / fer	Alarm; door loop
95			**TEST 6 – pass all before stage 7**						**You must see**
96	☐	6.1	dead Loose ends 166 ↔ 167 and 168 ↔ 169						**not 0 Ω**
97	☐	6.2	**LIVE** QF0 (main breaker), QF4 (24 V PSU + coil breaker) ON: loose ends 166 ↔ 167 and 168 ↔ 169. Then QF0 OFF, unplug, fit the A11 (controller) ends						**23–25 V DC, 166 / 168 positive; green and red lamps off**
98	☐	6.3	**LIVE** Power up						**handheld starts, fan runs**
99	☐	6.4	**LIVE** Handheld status						**E-stop / X7 (E-stop input) alarm is expected until stage 7**
100	**Stage 7 · E-stop input, sensors and tool setter – 17 wires, 12.70 m**								
101	⚠ **CHECK FIRST:** NPN NO sensors: brown +24 V, blue 0 V. Slide the GX16 (round connector) shell and heat-shrink on before soldering. Tool setter unplugged: GX16-5 dummy plug (pins 3–4 bridged) in XT (tool-setter socket).								
102	☐	**210**	E-stop 4	A11 X7	1 yellow	**dk blue**	**2.00**	fer / fer	E-stop signal to the controller; door loop ⚠ **§ 7.1 (dead tests): X7 (E-stop input) opens with the E-stop pushed**
103	☐	**330**	X-LIM 1	A11 X1	1 yellow	**dk blue**	**0.70**	solder / fer	X home (X–)
104	☐	**331**	Y-LIM 1	A11 X2	1 yellow	**dk blue**	**0.75**	solder / fer	Y home (Y–)
105	☐	**332**	Z-LIM 2	A11 X3	1 yellow	**dk blue**	**0.80**	solder / fer	Z home (Z+)
106	☐	**334**	X-LIM 2	Y-LIM 2	1 yellow	**dk blue**	**0.40**	solder / solder	Y+ limit (loop)
107	☐	**335**	Y-LIM 2	Z-LIM 1	1 yellow	**dk blue**	**0.40**	solder / solder	Z– limit (loop)
108	☐	**336**	X-LIM 2	A11 X6	1 yellow	**dk blue**	**0.75**	solder / fer	Hard limit (wire-OR)
109	☐	**337**	A11 GND	X-LIM 3	1 yellow	**bl/wh**	**0.75**	2×+169 / solder	Sensor 0 V (BU)

	A	B	C	D	E	F	G	H	I
4		No.	From	To	Wire mm²	Band	Cut	Ends (from / to)	Function · ⚠ check before / while wiring
110	☐	**338**	X-LIM 3	Y-LIM 3	1 yellow	**bl/wh**	**0.40**	solder / solder	Sensor 0 V (loop)
111	☐	**339**	Y-LIM 3	Z-LIM 3	1 yellow	**bl/wh**	**0.40**	solder / solder	Sensor 0 V (loop)
112	☐	**340**	A11 24V	X-LIM 4	1 yellow	**dk blue**	**0.75**	2×+168 / solder	Sensor +24 V (BN)
113	☐	**341**	X-LIM 4	Y-LIM 4	1 yellow	**dk blue**	**0.40**	solder / solder	Sensor +24 V (loop)
114	☐	**342**	Y-LIM 4	Z-LIM 4	1 yellow	**dk blue**	**0.40**	solder / solder	Sensor +24 V (loop)
115	☐	**345**	Tool setter 1	A11 X4	1 yellow	**dk blue**	**0.85**	solder / fer	Tool-setter touch
116	☐	**346**	Z-LIM 3	Tool setter 2	1 yellow	**bl/wh**	**0.40**	solder / solder	0 V (loop)
117	☐	**347**	Tool setter 3	E-stop 3	1 yellow	**dk blue**	**2.35**	solder / fer	Tool-setter over-travel (NC) in series with the E-stop NC2 → A11 X7 (door loop); door loop ⚠ **§ 7.1 (dead tests): X7 (E-stop input) opens with over-travel pushed**
118	☐	**348**	Tool setter 2	Tool setter 4	1 yellow	**bl/wh**	**0.20**	solder / solder	0 V to the over-travel switch (jumper in the connector)
119			**TEST 7 – pass all before stage 8**						**You must see**
120	☐	7.1	dead § 7.1 (dead tests): A11 (controller) X7 (E-stop input) ↔ G2 (24 V PSU) −V, E-stop and over-travel released						**beep**
121	☐	7.2	dead Same probes, E-stop pushed; then over-travel pushed						**open, open**
122	☐	7.3	dead X / Y / Z-LIM 4 ↔ A11 (controller) 24V; LIM 3 ↔ A11 GND; LIM 4 ↔ LIM 3						**beep; beep; not 0 Ω**
123	☐	7.4	**LIVE** Power up, E-stop released						**the E-stop / X7 (E-stop input) alarm clears**
124	☐	7.5	**LIVE** Hold steel at each sensor						**sensor LED on, its input changes: X1 / X2 / X3 home, X6 limit**
125	☐	7.6	**LIVE** Touch the tool-setter plate to the tool						**X4 (tool-setter input) changes**
126	**Stage 8 · 48 V supply, drives and motor cables – 9 wires, 3 cables, 11.90 m**								
127	⚠ **CHECK FIRST:** PSU input selector (if fitted) on 230 V. Fit the drive ends only after § 7.2 (power-up) reads G1 (48 V PSU) at 46–50 V DC. R3 open on 150 / 152 / 154 (1 mm² on the 15 A PSU): fuse each drive or use 1.5–2.5 mm² first. Set each drive's current and microstep switches for its motor before power. Motor cables: Ω-check the coil pairs first; never plug or unplug a motor with its drive powered.								
128	☐	**117**	L busbar XL	CB-48V 1	4 black		**0.60**	fer·own / fer	Switched L
129	☐	**118**	CB-48V 2	PSU 48V L	1 yellow	**black**	**1.15**	fer / fork	
130	☐	**119**	N busbar XN	PSU 48V N	1 yellow	**blue**	**0.95**	fer·own / fork	
131	☐	**150**	PSU 48V +V1	DMA860H-X AC1	1 yellow	**black**	**1.00**	fork / fer	48 V ⚠ **drive end off until G1 (48 V PSU) reads 46–50 V; R3: fuse or ≥ 1.5 mm²**
132	☐	**151**	PSU 48V -V1	DMA860H-X AC2	1 yellow	**black**	**0.95**	fork / fer	⚠ **drive end off until § 7.2 (power-up)**
133	☐	**152**	PSU 48V +V2	DMA860H-Y AC1	1 yellow	**black**	**1.10**	fork / fer	⚠ **drive end off until G1 (48 V PSU) reads 46–50 V; R3: fuse or ≥ 1.5 mm²**
134	☐	**153**	PSU 48V -V2	DMA860H-Y AC2	1 yellow	**black**	**1.05**	fork / fer	⚠ **drive end off until § 7.2 (power-up)**
135	☐	**154**	PSU 48V +V3	DMA860H-Z AC1	1 yellow	**black**	**1.00**	fork / fer	⚠ **drive end off until G1 (48 V PSU) reads 46–50 V; R3: fuse or ≥ 1.5 mm²**
136	☐	**155**	PSU 48V -V3	DMA860H-Z AC2	1 yellow	**black**	**1.00**	fork / fer	⚠ **drive end off until § 7.2 (power-up)**
137	☐		**Motor cable X: LiYCY 4 × 1.0 shielded**				**1.15**	DMA860H-X A+ → XM-X 1 · DMA860H-X A- → XM-X 2 · DMA860H-X B+ → XM-X 3 · DMA860H-X B- → XM-X 4	
138	☐		**Motor cable Y: LiYCY 4 × 1.0 shielded**				**1.00**	DMA860H-Y A+ → XM-Y 1 · DMA860H-Y A- → XM-Y 2 · DMA860H-Y B+ → XM-Y 3 · DMA860H-Y B- → XM-Y 4	
139	☐		**Motor cable Z: LiYCY 4 × 1.0 shielded**				**0.95**	DMA860H-Z A+ → XM-Z 1 · DMA860H-Z A- → XM-Z 2 · DMA860H-Z B+ → XM-Z 3 · DMA860H-Z B- → XM-Z 4	
140			**TEST 8 – pass all before stage 9**						**You must see**
141	☐	8.1	dead G1 (48 V PSU) +V ↔ G1 −V						**not 0 Ω**
142	☐	8.2	**LIVE** 150–155 loose at the drives; ON, QF2 (48 V PSU breaker) ON: G1 (48 V PSU) +V ↔ G1 −V. Then QF0 (main breaker) OFF, wait for the G1 LED to go out						**46–50 V DC (else adjust G1 (48 V PSU))**
143	☐	8.3	dead Each motor at the drive end: A+ ↔ A−, B+ ↔ B−; A+ ↔ B+						**a few Ω each; open**
144	☐	8.4	**LIVE** Fit 150–155; ON, QF2 (48 V PSU breaker) ON						**drive power LEDs on, no alarm LED, motors hold**
145	**Stage 9 · Step / direction signals and drive alarm – 18 wires, 13.70 m**								
146	⚠ **CHECK FIRST:** 5 V / PULSE / DIR are 5 V signals: never 24 V on them; twisted, away from 230 V and motor cables.								
147	☐	**301**	A11 X_5V	DMA860H-X PUL+	1 yellow tw	**dk blue**	**0.85**	fer / 2×+304	
148	☐	**302**	A11 X_PUL	DMA860H-X PUL-	1 yellow tw	**dk blue**	**0.85**	fer / fer	
149	☐	**303**	A11 X_DIR	DMA860H-X DIR-	1 yellow tw	**dk blue**	**0.95**	fer / fer	
150	☐	**304**	DMA860H-X PUL+	DMA860H-X DIR+	1 yellow	**dk blue**	**0.20**	2×+301 / fer	Jumper
151	☐	**305**	A11 Y_5V	DMA860H-Y PUL+	1 yellow tw	**dk blue**	**1.05**	fer / 2×+308	
152	☐	**306**	A11 Y_PUL	DMA860H-Y PUL-	1 yellow tw	**dk blue**	**1.05**	fer / fer	
153	☐	**307**	A11 Y_DIR	DMA860H-Y DIR-	1 yellow tw	**dk blue**	**1.05**	fer / fer	
154	☐	**308**	DMA860H-Y PUL+	DMA860H-Y DIR+	1 yellow	**dk blue**	**0.20**	2×+305 / fer	Jumper
155	☐	**309**	A11 Z_5V	DMA860H-Z PUL+	1 yellow tw	**dk blue**	**1.15**	fer / 2×+312	
156	☐	**310**	A11 Z_PUL	DMA860H-Z PUL-	1 yellow tw	**dk blue**	**1.25**	fer / fer	
157	☐	**311**	A11 Z_DIR	DMA860H-Z DIR-	1 yellow tw	**dk blue**	**1.25**	fer / fer	
158	☐	**312**	DMA860H-Z PUL+	DMA860H-Z DIR+	1 yellow	**dk blue**	**0.20**	2×+309 / fer	Jumper
159	☐	**315**	DMA860H-X ALM+	A11 X5	1 yellow	**dk blue**	**0.75**	2×+317 / fer	Drive alarm (wire-OR)
160	☐	**316**	A11 OGND	DMA860H-X ALM-	1 yellow	**bl/wh**	**0.75**	2×+326 / 2×+318	

		No.	From	To	Wire mm²	Band	Cut	Ends (from / to)	Function · ⚠ check before / while wiring
161	☐	**317**	DMA860H-Y ALM+	DMA860H-X ALM+	1 yellow	**dk blue**	**0.55**	2×+319 / 2×+315	NEW: X5 had 3 conductors
162	☐	**318**	DMA860H-X ALM-	DMA860H-Y ALM-	1 yellow	**bl/wh**	**0.55**	2×+316 / 2×+320	
163	☐	**319**	DMA860H-Z ALM+	DMA860H-Y ALM+	1 yellow	**dk blue**	**0.55**	fer / 2×+317	
164	☐	**320**	DMA860H-Y ALM-	DMA860H-Z ALM-	1 yellow	**bl/wh**	**0.50**	2×+318 / fer	
165			**TEST 9 – pass all before stage 10**						**You must see**
166	☐	9.1	dead § 7.1 (dead tests): A11 (controller) X / Y / Z 5V ↔ G2 (24 V PSU) +V; A11 PUL ↔ G1 (48 V PSU) +V						**open; open**
167	☐	9.2	dead Jumpers 304 / 308 / 312: PUL+ ↔ DIR+ at each drive						**beep**
168	☐	9.3	dead A11 (controller) X5 (drive alarm input) ↔ ALM+ of each drive; A11 OGND ↔ ALM− of each drive						**beep**
169	☐	9.4	**LIVE** Jog each axis slowly with the handheld						**smooth, the right way (else change the direction setting)**
170	☐	9.5	**LIVE** Home each axis at low speed						**stops at its home sensor**
171	**Stage 10 · VFD, braking resistor and spindle – 10 wires, 1 cable, 9.85 m**								
172	⚠ **CHECK FIRST:** VFD (spindle drive) L / N in, U / V / W out: never mains on U / V / W. Braking resistor tube ≥ 50 mm from ducts and wires. W1 (spindle cable): 360° shield at the HDC-HA (spindle connector) hood and the VFD clamp; U / V / W / PE only. Before the first cut, limit the VFD to ≈ 1.5 kW (C14: Z1 (EMI filter) is 20 A).								
173	☐	**114**	L busbar XL	CB-VFD 1	4 black		**0.60**	fer·own / fer	Switched L
174	☐	**115**	CB-VFD 2	VFD L	2.5 black		**1.00**	fer / fer	**⚠ VFD (spindle drive) input L, never U / V / W**
175	☐	**116**	N busbar XN	VFD N	2.5 blue		**1.25**	fer·own / fer	**⚠ VFD (spindle drive) input N, never U / V / W**
176	☐	**128**	VFD P+	Braking resistor T1	1 yellow	**black**	**1.25**	fer / ring	**⚠ keep ≥ 50 mm from the tube (very hot)**
177	☐	**129**	VFD PB	Braking resistor T2	1 yellow	**black**	**1.00**	fer / ring	**⚠ keep ≥ 50 mm from the tube (very hot)**
178	☐	**325**	A11 Y1	VFD X1	1 yellow	**dk blue**	**0.70**	fer / fer	Spindle run (forward)
179	☐	**326**	A11 OGND	VFD COM	1 yellow	**bl/wh**	**0.80**	2×+316 / fer	Output common **⚠ § 7.2 (power-up): VFD (spindle drive) COM ↔ G2 (24 V PSU) –V < 1 V DC**
180	☐	**327**	A11 Y2	VFD X2	1 yellow	**dk blue**	**0.70**	fer / fer	Multi-speed bit 1
181	☐	**328**	A11 Y3	VFD X3	1 yellow	**dk blue**	**0.75**	fer / fer	Multi-speed bit 2
182	☐	**329**	A11 Y4	VFD X4	1 yellow	**dk blue**	**0.75**	fer / fer	Multi-speed bit 3
183	☐		**Spindle cable W1: 4G1.5 shielded**				**1.05**	123: VFD U → HDC-HA 1 · 124: VFD V → HDC-HA 2 · 125: VFD W → HDC-HA 3 · 126: VFD PE → HDC-HA PE · 127: HDC-HA SH → VFD PEb	
184			**TEST 10 – pass all before stage 11**						**You must see**
185	☐	10.1	dead Braking resistor alone, before 128 / 129: T1 ↔ T2						**≈ 100 Ω**
186	☐	10.2	dead VFD (spindle drive) L ↔ N; spindle U / V / W ↔ PE at the HDC-HA (spindle connector)						**not 0 Ω; open**
187	☐	10.3	**LIVE** Push ON, QF1 (VFD breaker) ON						**VFD (spindle drive) display on**
188	☐	10.4	**LIVE** § 7.2 (power-up): VFD (spindle drive) X1 ↔ VFD COM; VFD COM ↔ G2 (24 V PSU) –V						**≤ 24 V DC, X1 positive; < 1 V DC**
189	☐	10.5	**LIVE** Start the spindle slowly from the handheld; try each speed grade						**turns the right way, each grade its own frequency**
190	☐	10.6	**LIVE** Heavy test cut, clamp meter on Z1 (EMI filter) LINE L						**< 20 A (else lower the VFD limit)**
191	**Stage 11 · Final check**								
192	⚠ **CHECK FIRST:** All ducts covered, every screw tug-tested, every number sleeve and colour band on.								
193			**TEST 11 – pass all before the first job**						**You must see**
194	☐	11.1	dead Repeat the whole § 7.1 (dead tests) dead test (simulator: Procedures, P)						**all pass**
195	☐	11.2	**LIVE** Repeat the § 7.2 (power-up) power-up in order						**all pass**
196	☐	11.3	dead Power off: open and close the door 10 times, repeat tests 3.1–3.3						**unchanged**
197	Not listed: 12 conductors outside the box (motor and sensor cables to the machine) · 95.2 m of single wire to cut in all · source A11_Wire_Lengths_rev07.csv, regenerate with make_build_steps.py								